

URL, DNS, HTTP & HTTPS

Lesson 35 — Addresses and Secure Transfer

JKSSB Computer Awareness

October 8, 2026

Why Addresses Matter on the Internet

- ▶ Every computer on the Internet is reached by a numeric **IP address**, but people type readable names instead.
- ▶ A **URL** names one exact page or file; **DNS** translates the readable name into the numeric address.
- ▶ **HTTP** carries web pages between browser and server; **HTTPS** is HTTP protected by **TLS**.
- ▶ This lesson follows one typed address all the way to a secure page: URL → DNS → IP → server.

The one idea

Names are for people; numbers are for machines; DNS connects the two, and HTTPS protects the trip.

What a URL Is

- ▶ **URL** stands for Uniform Resource Locator.
- ▶ It is the full **address of one resource**: one page, file, or service on the Internet.
- ▶ Every URL has two halves: **where and how** (scheme plus host) and **exactly what** (port, path, and query).
- ▶ Example used in this lesson:
`https://www.example.in:443/study/computer?topic=dns&lang=en`

Keep the pair straight

The URL addresses one resource; the domain name is only the host part inside it.

URL Anatomy at a Glance

Part	Example piece	Role
Scheme	https	Protocol to use
Subdomain	www	Section of the site
Domain + TLD	example.in	Registered name
Port	443	Door number on the server
Path	/study/computer	Location of the page
Query	?topic=dns	Extra parameters

Learn the order: scheme, host, optional port, path, then query.

Scheme: How to Connect

- ▶ The **scheme** is the part before the colon: `https`, `http`, `ftp`, or `mailto`.
- ▶ It tells the browser **which protocol** to speak to the server.
- ▶ **http** means web transfer without built-in security; **https** means the same transfer protected by TLS.
- ▶ The `://` after the scheme simply separates the scheme from the host name.

Trap alert

The scheme is the protocol word (`https`), not the domain name that follows it.

Domain, Subdomain, and TLD

- ▶ The **domain name** is the readable host, such as `www.example.in`.
- ▶ The **TLD** (Top-Level Domain) is the last label: `.in`, `.com`, `.org`, `.edu`, `.gov`.
- ▶ The **subdomain** is a prefix section: `www`, `mail`, or `blog` in front of the registered domain.
- ▶ One registered domain can carry many subdomains, each pointing to a different service.

Worked split

In `mail.example.com`: `mail` is the subdomain, `example` is the domain, and `.com` is the TLD.

Port, Path, and Query

- ▶ The **port** is the door number on the server: 80 is the default for HTTP, **443** is the default for HTTPS.
- ▶ The **path** locates the page inside the site, such as `/study/computer`.
- ▶ The **query** passes parameters after a `?`, such as `?topic=dns&lang=en`.
- ▶ Ports and queries are **optional**: browsers fill in the default port when it is missing.

Defaults to remember

HTTP assumes port 80; HTTPS assumes port 443, so both are usually hidden in the address bar.

IP Address: The Real Destination

- ▶ An **IP** (Internet Protocol) address is the numeric address of a device on the network.
- ▶ **IPv4** is 32-bit, written as four decimal groups, for example 192.0.2.44.
- ▶ **IPv6** is 128-bit, written in hexadecimal groups; it is never 256-bit.
- ▶ Servers, routers, and the office desktop each need an IP address to send and receive data.

Trap alert

IPv6 is **128-bit**, not 256-bit — a repeated wrong option.

DNS: The Phone Book of the Internet

- ▶ **DNS** stands for Domain Name System.
- ▶ It translates **domain names** (typed by people) into **IP addresses** (used by machines).
- ▶ DNS is a distributed, hierarchical system: resolvers ask servers level by level until the answer is found.
- ▶ Results are **cached**, so a repeated visit skips most of the lookup steps.

Definition

DNS is the naming system that maps each domain name to the IP address of its server.

DNS Resolution Step by Step

- ▶ The browser first checks its own **cache**: is this name already known?
- ▶ If not, the **resolver** (usually run by the ISP) asks the DNS hierarchy: root, then TLD, then the domain's own server.
- ▶ The domain's server returns the **authoritative** IP address for that name.
- ▶ The resolver caches the answer and hands the IP address back to the browser.

Office desktop trace

Typing `www.example.in` asks DNS for its IP (say `192.0.2.44`); only then can the browser open the connection.

HTTP: Carrying Web Pages

- ▶ **HTTP** stands for Hypertext Transfer Protocol.
- ▶ It is the **request–response** protocol of the web: the browser requests a page, the server returns it.
- ▶ HTTP runs by default over **port 80**.
- ▶ Plain HTTP does **not** encrypt the data, so anyone on the path can read it.

Role in one line

HTTP moves web content; it does not protect it.

HTTPS and TLS: HTTP Made Secure

- ▶ **HTTPS** stands for Hypertext Transfer Protocol Secure.
- ▶ It is **HTTP protected by TLS**: TLS (Transport Layer Security) encrypts the whole exchange.
- ▶ HTTPS runs by default over **port 443** and shows the padlock in the address bar.
- ▶ TLS gives **confidentiality** (encryption), **integrity** (tamper detection), and server **authentication** (certificates).

Trap alert

HTTPS is HTTP **over TLS**; TLS replaced the older SSL, whose name survives only in the phrase “SSL/TLS”.

HTTP vs HTTPS

Point	HTTP	HTTPS
Security	No encryption	Encrypted by TLS
Default port	80	443
Address bar	http://	https:// + padlock

Trap alert

Port 80 belongs to HTTP; port 443 belongs to HTTPS. Reversing them is a common wrong answer.

The Full Flow: URL to Secure Page

- ▶ Type the **URL**; the browser reads the scheme, host, port, path, and query.
- ▶ **DNS** resolves the host name to an **IP address**.
- ▶ The browser opens a connection to that IP on the right **port** (80 or 443).
- ▶ Over **HTTPS**, a TLS handshake secures the channel; then the HTTP request for the **path** travels inside it.

The chain

URL → DNS → IP → server → TLS (if https) → page.

Trap Alert: Facts to Keep Straight

- ▶ URL addresses **one resource**; the domain is only its host part.
- ▶ DNS maps names to IPs; it does **not** assign addresses or carry page content.
- ▶ IPv4 is **32-bit**; IPv6 is **128-bit**, never 256-bit.
- ▶ HTTP uses port **80**; HTTPS uses port **443**.
- ▶ HTTPS is HTTP over **TLS**, not a different language; HTTP alone gives no encryption.

Lesson 35: Recall

- ▶ URL parts in order: scheme, subdomain, domain, TLD, port, path, query.
- ▶ URL stands for Uniform Resource Locator; DNS stands for Domain Name System; HTTP stands for Hypertext Transfer Protocol; TLS stands for Transport Layer Security.
- ▶ DNS translates domain names into IP addresses through cached, hierarchical lookups.
- ▶ IPv4 is 32-bit; IPv6 is 128-bit.
- ▶ HTTP (port 80) carries pages; HTTPS (port 443) is HTTP protected by TLS.
- ▶ The flow is URL → DNS → IP → server → secure page.